

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

SESSION: 2025-26



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Game Scoreboard System (Using Core Java & OOP Concepts)

Project Description

The Game Scoreboard System is a console-based Java application that manages player scores and displays rankings. It stores player details, sorts them based on performance, and generates a leaderboard.

Objectives

- Design player management system
- Store and manage scores
- Use arrays of objects
- Implement sorting logic
- Display leaderboard

Approach

1. Create Player class
2. Use constructor
3. Store players in array
4. Apply Bubble Sort
5. Display leaderboard

Java Code

```
class Player {
    private String playerName;
    private int score;

    public Player(String playerName, int score) {
        this.playerName = playerName;
        this.score = score;
    }

    public String getName() { return playerName; }
    public int getScore() { return score; }
}

public class GameSystem {
    public static void main(String[] args) {
        Player[] players = {
            new Player("Rahul", 95),
            new Player("Ankit", 88),
            new Player("Priya", 82),
            new Player("Neha", 76)
        };

        for (int i = 0; i < players.length - 1; i++) {
            for (int j = 0; j < players.length - i - 1; j++) {
                if (players[j].getScore() < players[j + 1].getScore()) {
                    Player temp = players[j];
                    players[j] = players[j + 1];
                    players[j + 1] = temp;
                }
            }
        }
    }
}
```

```
        }  
    }  
  
    System.out.println("Game Scoreboard");  
    for (int i = 0; i < players.length; i++) {  
        System.out.println((i+1) + " " + players[i].getName() + " " + players[i].getScore());  
    }  
}  
}
```

Sample Output

Game Scoreboard
1 Rahul 95
2 Ankit 88
3 Priya 82
4 Neha 76
Top Player: Rahul

Conclusion

This project demonstrates Java OOP concepts like classes, objects, arrays, and sorting in a real-world application.